PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Agentic AI–Driven Intelligent RPA Task Orchestrator for Automated Document Processing

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Title of the Project"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Swati Goel The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Rishabh Jain**

Date: 18 May 2026 (2210990725)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who contributed to the successful completion of this CO-OP Project titled **“Agentic AI–Driven Intelligent RPA Task Orchestrator for Automated Document Processing.”**

First and foremost, I would like to thank **Chitkara University, Punjab**, for providing the opportunity, resources, and academic environment required to undertake and complete this project successfully.

I extend my heartfelt gratitude to my project mentor and supervisor, **Swati Goel,** for their continuous guidance, encouragement, and valuable suggestions throughout the development of this project. Their support and technical insights played an important role in shaping the direction and successful implementation of the work.

I would also like to thank the faculty members of the **Department of Computer Science and Engineering** for their constant motivation and support during the course of this project. Special thanks to my friends and peers who provided valuable feedback, discussions, and encouragement during various stages of the project.

Finally, I would like to express my gratitude to my family for their continuous support, motivation, and encouragement throughout my academic journey and during the completion of this project.

The experience gained while working on this project has greatly enhanced my understanding of intelligent automation, robotic process automation, OCR technologies, and workflow orchestration systems.

Abstract

Traditional Robotic Process Automation (RPA) systems operate on static and predefined workflows, which limits their ability to adapt to changing business requirements and unstructured data inputs. Organizations frequently receive documents such as invoices, reports, and forms through emails in varying formats, requiring manual classification and workflow selection before processing. This dependence on human intervention reduces efficiency, scalability, and flexibility in enterprise automation systems.

To address these limitations, this project proposes an Agentic AI–Driven Intelligent RPA Task Orchestrator for Automated Document Processing. The proposed system integrates an agentic decision-making layer with traditional RPA workflows to enable adaptive and context-aware document processing. Instead of relying solely on fixed workflows, the system dynamically interprets document content and determines suitable processing actions automatically.

The implemented framework monitors an email inbox for incoming document attachments, downloads PDF files, and extracts textual information using OCR technologies such as Microsoft OCR or Tesseract OCR. The extracted text is processed by a Python-based agentic classification module that analyzes document content and determines the document category and processing strategy. Based on the classification result, the UiPath orchestration layer dynamically routes and processes the documents. Logging and exception handling mechanisms are incorporated to ensure transparency, traceability, and system reliability.

The system follows a modular architecture consisting of email ingestion, attachment extraction, OCR preprocessing, agentic decision-making, workflow orchestration, and logging modules. This modular design improves maintainability, scalability, and extensibility of the solution.

The developed prototype demonstrates how an intelligent agent-based orchestration layer can enhance traditional RPA systems by enabling adaptive automation and reducing manual intervention. The system successfully automates document ingestion, OCR-based text extraction, dynamic routing, and execution logging. The project establishes a strong foundation for future enterprise-level intelligent automation systems capable of handling unstructured data efficiently.

1. Introduction

Robotic Process Automation (RPA) has emerged as one of the most widely adopted technologies for automating repetitive and rule-based business operations. Organizations use RPA systems to automate tasks such as data entry, invoice processing, report generation, and email handling. Traditional RPA solutions improve operational efficiency by reducing human effort and increasing process speed.

However, conventional RPA systems operate on static workflows and predefined execution logic. Such systems perform efficiently only when the structure of inputs and processing requirements remain unchanged. In real-world enterprise environments, organizations frequently receive unstructured and semi-structured documents in varying formats through emails and other communication channels. Processing such documents requires dynamic decision-making capabilities that traditional RPA systems cannot provide effectively.

Manual intervention is often required for identifying document types, selecting appropriate workflows, and handling processing exceptions. This dependency limits the scalability and flexibility of existing automation systems. As businesses increasingly rely on digital document workflows, there is a growing need for intelligent automation systems capable of adapting dynamically to varying document structures and processing requirements.

This project introduces an Agentic AI–Driven Intelligent RPA Task Orchestrator for Automated Document Processing. The proposed solution integrates an intelligent agent-based decision layer with traditional RPA workflows. The system automatically monitors email inboxes, extracts document attachments, performs OCR-based text extraction, and dynamically routes documents using a Python-based classification engine.

The proposed framework enables adaptive workflow orchestration and reduces manual dependency in document processing systems. The architecture follows a modular approach that separates ingestion, preprocessing, decision-making, orchestration, and logging functionalities, thereby improving maintainability and scalability.

2. Methodology

2.1 Email-Based Document Ingestion

The system monitors an Outlook inbox using UiPath automation workflows. Emails matching predefined criteria are fetched automatically, and unread messages are processed in batches.

2.2 Attachment Extraction

PDF attachments are extracted from emails using UiPath activities and stored in dynamically generated folders using sender details and timestamps.

2.3 OCR-Based Text Extraction

The extracted PDF documents are processed using offline OCR engines such as Microsoft OCR or Tesseract OCR. OCR converts scanned and unstructured documents into machine-readable text.

2.4 Agentic Decision Engine

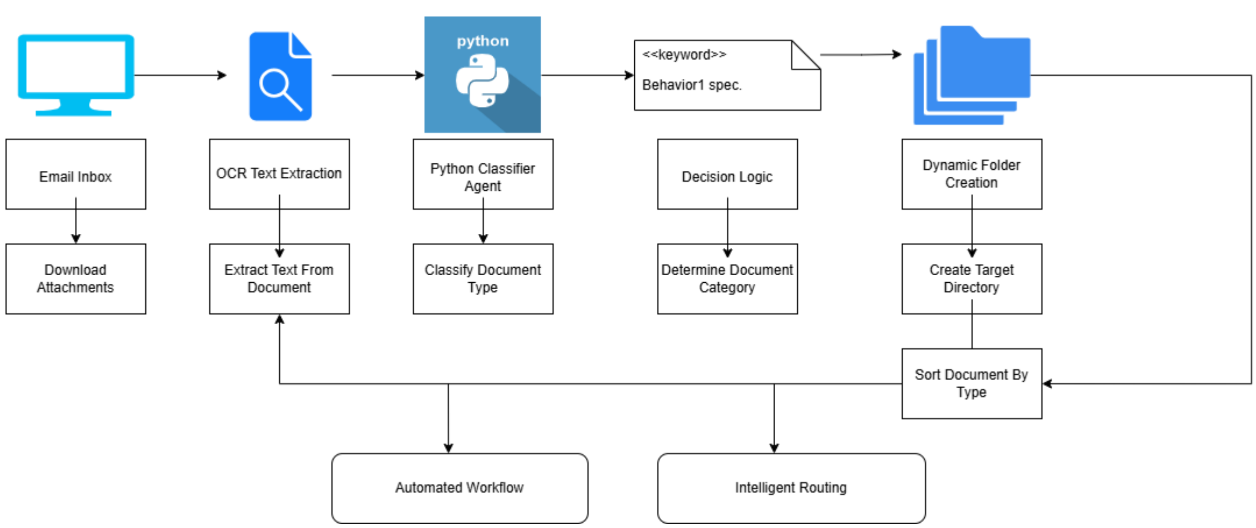
A Python-based document classifier analyzes OCR-generated text using keyword-based scoring and contextual analysis techniques. The system determines document categories dynamically.

2.5 Dynamic Workflow Orchestration

Based on the classification result, the UiPath orchestration layer dynamically routes documents and executes suitable processing workflows.

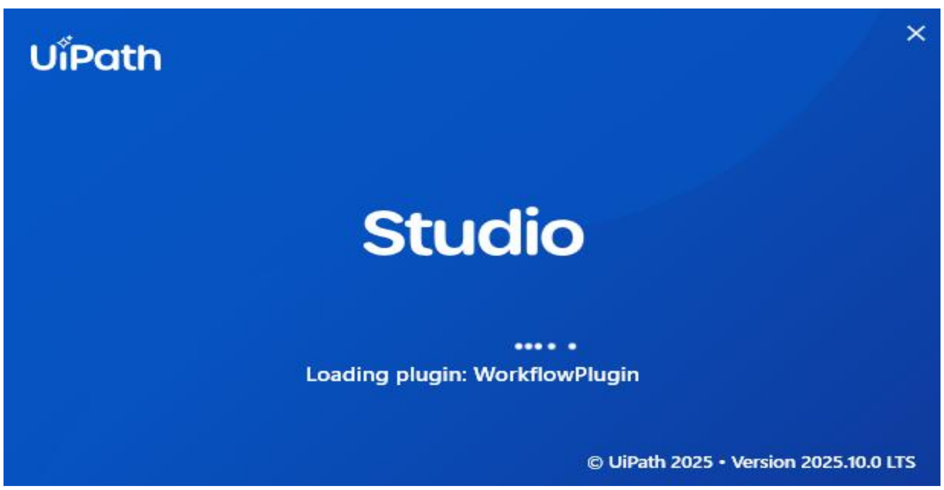
2.6 Logging and Monitoring

All execution activities, processing details, and exceptions are logged to ensure traceability, debugging support, and transparency.



3. TOOLS AND TECHNOLOGIES

Tool / Technology	Purpose
UiPath	Workflow automation and orchestration
Python	Agentic document classification
Outlook Integration	Email monitoring and retrieval
Microsoft OCR / Tesseract OCR	Text extraction from documents
File System Routing	Dynamic document organization
TXT / JSON Communication	Data exchange between modules



4. IMPLEMENTATION

4.1 Email Monitoring Workflow

The system uses UiPath's Get Outlook Mail Messages activity to monitor unread emails containing document attachments. Relevant emails are filtered using subject-based conditions.

4.2 Attachment Processing

Attachments are downloaded automatically and organized into timestamp-based folders to maintain uniqueness and traceability.

4.3 OCR Processing Pipeline

The downloaded PDF files are processed individually using OCR engines. Extracted text is stored as .txt files for downstream processing.

4.4 Python-Based Classification

The agentic decision engine analyses extracted text and determines document type using keyword-based classification techniques.

4.5 Dynamic Routing

The UiPath workflow dynamically routes documents and performs suitable actions based on the classification output generated by the Python classifier.

4.6 Logging and Exception Handling

Try-Catch mechanisms and logging activities ensure fault tolerance and execution transparency throughout the workflow.

3. Result

5.1 Results Achieved

The developed system successfully demonstrates an intelligent and event-driven document processing pipeline.

The following functionalities were successfully implemented:

- Automated email monitoring
- PDF attachment extraction
- OCR-based text extraction
- Raw text generation
- Dynamic document handling
- Logging and error handling

5.2 System Outputs

The system generates:

- OCR-extracted text files
- Execution logs
- Organized document storage folders

5.3 System Characteristics

- Event-driven architecture
- Modular workflow design
- Fault-tolerant execution
- Scalable processing pipeline

- Dynamic workflow orchestration

5.4 Benefits Achieved

- Reduction in manual effort
- Improved processing efficiency
- Enhanced automation flexibility
- Better traceability and monitoring
- Ability to process unstructured documents

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project successfully demonstrates the implementation of an intelligent RPA orchestration framework for automated document processing. The proposed solution integrates an agentic decision-making layer with traditional RPA workflows to enable adaptive and context-aware automation.

The system automates email monitoring, attachment extraction, OCR preprocessing, document classification, and workflow routing. The modular architecture improves scalability, maintainability, and reliability of the automation pipeline.

The developed prototype establishes a strong foundation for intelligent enterprise automation systems capable of processing unstructured data efficiently with minimal human intervention.

6.2 Future Scope

The future scope of the project includes:

- Support for multiple document types
- Integration of advanced AI and machine learning models
- Dashboard-based monitoring and analytics
- Human-in-the-loop approval mechanisms
- Cloud-based deployment
- Integration with enterprise ERP and CRM systems
- Confidence-based workflow orchestration

REFERENCES

1. UiPath Documentation – RPA Workflow Automation
2. Python Official Documentation
3. Microsoft OCR Documentation
4. Tesseract OCR Documentation
5. Research papers on Intelligent Document Processing and RPA
6. Enterprise Automation and Workflow Orchestration Studies